# US Patent & Trademark Office
# Patent Public Search | Text View


| | |
|---|---|
| United States Patent Application Publication | 20250273497 |
| Kind Code | A1 |
| Publication Date | August 28, 2025 |
| Inventor(s) | ELUMALAI; KARTHIK et al. |


## CATHODE ASSEMBLY FOR INTEGRATION OF EMBEDDED ELECTROSTATIC CHUCK (ESC)


### Abstract

Embodiments disclosed herein include an apparatus that includes an electrostatic chuck (ESC). The electrostatic chuck may include a first body that is electrically conductive, and a ceramic insert on the first body with an electrode embedded within the ceramic insert. In an embodiment, the apparatus may further include a facility plate that is coupled to the ESC. The facility plate may include a second body that is electrically conductive with a hole through the second body. In an embodiment, a DC input connector is provided through the hole, and an RF feed line is coupled to the second body. In an embodiment, a pin of the DC input connector is electrically isolated from the RF feed line.



**Inventors:** **ELUMALAI; KARTHIK (Singapore, SG), TATTI; ARUNKUMAR (Haveri, IN), MOHAMED YUNOS; MUHAMMAD DANIAL BIN (Singapore, SG), PEH; ENG SHENG (Singapore, SG), SUN; CHENG (Singapore, SG), LIU; YE Y (Singapore, SG)**

**Applicant:** **Applied Materials, Inc.** (Santa Clara, CA)



**Family ID:** **1000007842036**

**Appl. No.:** **18/586307**

**Filed:** **February 23, 2024**


## Publication Classification


**Int. Cl.:** **H01L21/683** (20060101); **H01J37/32** (20060101)

**U.S. Cl.:**

CPC **H01L21/6833** (20130101); **H01J37/32577** (20130101); **H01J37/32724** (20130101);

## Background/Summary

BACKGROUND
1) Field
[0001] Embodiments relate to the field of semiconductor manufacturing and, in particular, to cathode assemblies with an RF feed that is electrically isolated from a DC voltage supply.
2) Description of Related Art
[0002] In semiconductor manufacturing processes, the wafer (e.g., a silicon wafer) or other substrate is coupled to a pedestal during various processes. For example, the wafer may be secured to the pedestal during plasma dicing processes, etching processes, deposition processes, and/or the like. In some embodiments, the wafer is secured to the pedestal through the use of an electrostatic chuck (ESC). The ESC provides an electrostatic force that attracts the wafer to the chuck in order to prevent movement of the wafer during processing.
[0003] Typically, the ESC is fed several inputs from an underlying cathode assembly. In existing solutions, an RF feed (for applying an RF bias) and a DC supply (for generating a chucking force) are provided to the ESC through a single input. The coupling of the RF feed and the DC supply can lead to high current leakage. For example, electrical current from the DC supply can leak into the plasma environment.
SUMMARY
[0004] Embodiments disclosed herein include an apparatus that includes an electrostatic chuck (ESC). The electrostatic chuck may include a first body that is electrically conductive, and a ceramic insert on the first body with an electrode embedded within the ceramic insert. In an embodiment, the apparatus may further include a facility plate that is coupled to the ESC. The facility plate may include a second body that is electrically conductive with a hole through the second body. In an embodiment, a DC input connector is provided through the hole, and an RF feed line is coupled to the second body. In an embodiment, a pin of the DC input connector is electrically isolated from the RF feed line.
[0005] Embodiments may further include an apparatus that includes a body with a first surface and a second surface opposite from the first surface. The body may be electrically conductive. In an embodiment, a hole is provided through the body, and an input connector is inserted through the hole. In an embodiment, the input connector comprises a pin that is electrically conductive and a collar around the pin that is electrically insulating. In an embodiment, the apparatus further comprises an electrically conductive feed line contacting the body.
[0006] Embodiments may also comprise a semiconductor processing tool that includes a chamber suitable for maintaining a vacuum environment within the chamber. In an embodiment, the tool may also include a cathode assembly within the chamber. The cathode assembly may include an electrostatic chuck (ESC) and a facility plate. In an embodiment, the ESC includes a first body that is electrically conductive, and a ceramic insert on the first body, with an electrode embedded within the ceramic insert. In an embodiment, the facility plate is coupled to the ESC, and includes a second body that is electrically conductive, a hole through the second body, a DC input connector through the hole, and an RF feed line coupled to the second body. In an embodiment, a pin of the DC input connector is electrically isolated from the RF feed line.

---

## Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0007] FIG. **1**A is a perspective view illustration of a cathode assembly, in accordance with an embodiment.

[0008] FIG. **1**B is a cross-sectional illustration of a cathode assembly with a single input for the RF feed and the DC input, in accordance with an embodiment.

[0009] FIG. **2**A is a cross-sectional illustration of a facility plate with a DC input that is electrically isolated from an RF feed, in accordance with an embodiment.

[0010] FIG. **2**B is a cross-sectional illustration of a cathode assembly with an electrostatic chuck (ESC) and a facility plate, in accordance with an embodiment.

[0011] FIG. **3**A is a zoomed in cross-sectional illustration of the DC input that is electrically isolated from the metallic body of the ESC, in accordance with an embodiment.

[0012] FIG. **3**B is a cross-sectional illustration of a cathode assembly with an ESC and a facility plate with a DC input that is electrically isolated from an RF feed, in accordance with an embodiment.

[0013] FIGS. **4**A-**4**E are illustrations depicting a process for relocating a gas pathway through a facility plate, in accordance with an embodiment.

[0014] FIG. **4**F is a process flow diagram of a process for forming a gas pathway through a facility plate, in accordance with an embodiment.

[0015] FIG. **5** is cross-sectional illustration of a semiconductor processing tool that comprises a cathode assembly with a DC input that is electrically isolated from an RF feed, in accordance with an embodiment.

[0016] FIG. **6** is a process flow diagram of a method for processing a substrate with a cathode assembly that comprises a DC input that is electrically isolated from an RF feed, in accordance with an embodiment.

[0017] FIG. **7** illustrates a block diagram of an exemplary computer system that may be used in conjunction with a processing tool, in accordance with an embodiment.

## DETAILED DESCRIPTION

[0018] Embodiments described herein include apparatuses and methods for using a cathode assembly with an RF feed that is electrically isolated from a DC voltage supply. In the following description, numerous specific details are set forth in order to provide a thorough understanding of embodiments. It will be apparent to one skilled in the art that embodiments may be practiced without these specific details. In other instances, well-known aspects are not described in detail in order to not unnecessarily obscure embodiments. Furthermore, it is to be understood that the various embodiments shown in the accompanying drawings are illustrative representations and are not necessarily drawn to scale.

[0019] Various embodiments or aspects of the disclosure are described herein. In some implementations, the different embodiments are practiced separately. However, embodiments are not limited to embodiments being practiced in isolation. For example, two or more different embodiments can be combined together in order to be practiced as a single device, process, structure, or the like. The entirety of various embodiments can be combined together in some instances. In other instances, portions of a first embodiment can be combined with portions of one or more different embodiments. For example, a portion of a first embodiment can be combined with a portion of a second embodiment, or a portion of a first embodiment can be combined with a portion of a second embodiment and a portion of a third embodiment.

[0020] The embodiments illustrated and discussed in relation to the figures included herein are provided for the purpose of explaining some of the basic principles of the disclosure. However, the scope of this disclosure covers all related, potential, and/or possible, embodiments, even those differing from the idealized and/or illustrative examples presented. This disclosure covers even those embodiments which incorporate and/or utilize modern, future, and/or as of the time of this writing unknown, components, devices, systems, etc., as replacements for the functionally equivalent, analogous, and/or similar, components, devices, systems, etc., used in the embodiments illustrated and/or discussed herein for the purpose of explanation, illustration, and example.

[0021] As noted above, existing electrostatic chuck (ESC) structures are susceptible to significant current leakage. That leakage may result from the current (e.g., DC or RF) leaking away from the ESC and coupling with the plasma within a chamber and/or otherwise escaping through other electrical pathways. One problem with leakage is that chucking force is difficult to control. In some instances, current leakage during a processing operation within the chamber may result in dechucking of the wafer. That is, the chucking force dips below a given threshold, and the wafer becomes free to move, bend, or otherwise displace relative to the surface of the ESC. This movement is detrimental because the processing conditions are highly tuned, and the movement may result in non-uniform treatment of the wafer or damage to the wafer or ESC.

[0022] To prevent dechucking, the wafer is often "overchucked" so that the chucking force is increased beyond what would otherwise be necessary. This can lead to damage to the wafer and/or damage to the ESC itself. For example, an overchucked wafer can be cracked, chipped, deformed, or otherwise damaged. Overchucking may also lead to excessive wear on the ESC. This may require more frequent repair, replacement, and/or refurbishing of the ESC. As such, the leakage current can lead to increases in cost of ownership of a semiconductor processing tool, and/or an increase in manufacturing costs due to damaged wafers. The presence of leakage current also reduces the efficiency of the system. That is, more energy is needed in order to run a given process when overchucking is necessary to account for leakage. This also increases costs and can generate environmental impact issues.

[0023] Referring now to FIG. **1**A, a perspective view illustration of a cathode assembly **150** is shown, in accordance with an embodiment. In an embodiment, the cathode assembly **150** may comprise a facility plate **120** and an ESC **100**. The facility plate **120** is an interface layer that is used to mechanically couple the ESC **100** to the rest of the chamber or tool (not shown in FIG. **1**A). The facility plate **120** can function as an adapter to enable electrical coupling to the ESC **100** and/or to enable fluidic coupling (e.g., for backside gas flow).

[0024] In an embodiment, the ESC **100** may comprise a first body **101**. The body **101** may be an electrically conductive material. For example, the first body **101** may comprise aluminum, or the like. The first body **101** may have a first surface **103** (e.g., a bottom surface) and a second surface **104** (e.g., a top surface) opposite from the bottom surface. In an embodiment, the body has a cylindrical shape that is suitable for supporting a wafer, such as a standard silicon wafer. For example, a diameter of the first body **101** may be at least 200 mm or larger, at least 300 mm or larger, at least 450 mm or larger, or at least 750 mm or larger.

[0025] In an embodiment, a ceramic plate **105** may be provided on the second surface **104** of the body **101**. The ceramic plate **105** may be set into a recess of the second surface **104** so that the top surface of the ceramic plate **105** is substantially coplanar with a top surface of the first body **101**. Though, in other embodiments, the ceramic plate **105** may have a top surface above or below the top surface of the first body **101**.

[0026] In an embodiment, the ceramic plate **105** may comprise an electrically conductive electrode (not visible in FIG. **1**A). The electrode may be embedded within a thickness of the ceramic plate **105**. The electrode is coupled to a DC input in order to generate a chucking force that attracts and secures a wafer (not shown) to the ceramic plate **105**. In an embodiment, the ceramic plate may comprise any suitable material, such as aluminum nitride, aluminum oxide, or the like.

[0027] In an embodiment, the facility plate **120** may comprise a second body **122**. The second body **122** may also be electrically conductive. The second body **122** may be the same material as the first body **101**. For example, the second body **122** may comprise aluminum or the like. In some embodiments, a link **123** may be provided between the first body **101** and the second body **122**. The link **123** may be a bolt, a screw, a pin, or the like. In some embodiments, the link **123** is electrically conductive. As such, the link **123** may provide electrical coupling between the facility plate **120** and the ESC **100**. The link **123** may also provide mechanical coupling between the facility plate **120** and the ESC **100**.

[0028] In an embodiment, the facility plate **120** may have a diameter that is smaller than a diameter of the ESC **100**. Though, in other embodiments, the facility plate **120** may have a diameter that is substantially equal to a diameter of the ESC **100**, or the facility plate **120** may have a diameter that is greater than a diameter of the ESC **100**.

[0029] Referring now to FIG. **1**B, a cross-sectional illustration of the cathode assembly **150** in FIG. **1**A is shown, in accordance with an embodiment. As shown, the ceramic plate **105** in the ESC **100** is set into a recess along the second surface **104** of the body **101**. The ceramic plate **105** may also comprise an electrode **108**. The electrode **108** may be an electrically conductive material, such as copper or the like. The electrode **108** may be a conductive plate, a conductive mesh, or have any other conductive pattern that is distributed though the ceramic plate **105**. In the illustrated embodiment, the electrode **108** is set at a midpoint along a thickness direction of the ceramic plate **105**. Though, in other embodiments the electrode **108** may be closer to a top of the ceramic plate **105** or closer to a bottom of the ceramic plate **105**. When the electrode **108** is embedded within the ceramic plate **105** as shown in FIG. **1**B, the ESC **100** may sometimes be referred to as an embedded ESC **100**.

[0030] In an embodiment, the electrode **108** may be electrically coupled to an input **112**. For example, the input **112** may be in electrical contact with the second body **122** of the facility plate **120**, and the links **123** may electrically couple the second body **122** to the first body **101** of the ESC **100**. The first body **101** may be electrically coupled to the electrode **108** through a via or other contact (not shown) that passes through the ceramic plate **105**. In this way, a DC bias may be applied from the input **112** to the electrode **108** in order to provide a chucking force.

[0031] At the same time, the input **112** may also receive an RF signal that is transferred to the second body **122** of the facility plate **120** and the first body **101** of the ESC **100** (through the links **123**). This allows for an RF bias to be applied to the ESC **100** which may be used during processing operations. Since the DC input and the RF feed are not electrically isolated from each other, leakage is more prevalent in the cathode assembly **150** compared to embodiments that will be described in greater detail herein. This may lead to issues with dechucking, excessive energy consumption, and/or the like.

[0032] In an embodiment, the ESC **100** may also comprise fluidic channels **115**. The fluidic channels **115** may be provided at a bottom of the ESC **100** and sealed with a lid **116**. The fluidic channels **115** may be suitable for flowing gas and/or liquid within the body **101**. This can be used for cooling or other thermal control of the ESC **100**. In the illustrated embodiment, a single gas input **153** is coupled to the ESC **100** in order to distribute gas into the fluidic channels **115**.

[0033] As described above, the cathode assembly **150** in FIGS. **1**A and **1**B suffers from current leakage due to the combined input **112** for the DC current and RF current. Accordingly, embodiments disclosed herein may include an ESC that has a DC input line that is separate from the RF feed. More particularly, the DC voltage applied to the electrode within the ceramic plate can be electrically isolated from the conductive bodies of the facility plate and the ESC. Similarly, the RF feed can be provided directly to the conductive body of the ESC without being superimposed on the DC input line. Due to the electrical isolation, the leakage current can be reduced.

[0034] Additionally, embodiments disclosed herein provide enhanced thermal control through the use of a dual zone gas delivery system. In an embodiment, the gas delivery through the facility plate is designed in order to accommodate the gas input locations of the ESC. That is, the ESC does not require any redesign in order to route a dual gas zone solution to the ESC.

[0035] Referring now to FIG. **2**A, a cross-sectional illustration of a facility plate **220** is shown, in accordance with an embodiment. In an embodiment, the facility plate **220** may comprise an electrically conductive body **222**. The electrically conductive body **222** may comprise aluminum or another metallic material. In an embodiment, a hole **224** may pass through a thickness of the facility plate **220**. The hole **224** may be provided at an approximate center of the facility plate **220**. Though, the hole **224** may be provided at other locations, depending on the overall design of the cathode assembly. The hole **224** may be sized to receive a DC input connector. The DC input connector may be a high voltage connector that will ultimately be electrically coupled to the electrode of the ESC (not shown in FIG. **2**B). In an embodiment, the DC input connector may comprise a pin **225** that is electrically conductive. The DC input connector may further comprise an electrically insulating collar **226**. The electrically insulating collar **226** electrically isolates the pin **225** from the electrically conductive body **222** of the facility plate **220**.

[0036] In an embodiment, the facility plate **220** may be coupled to an RF feed line **221**. The RF feed line **221** may directly contact the body **222**. In other instances, an intervening electrically conductive structure may be provided between the RF feed line **221** and the body **222**. Electrical isolation between the pin **225** and the body **222** (which may be provided by collar **226**) allows for the body **222** to be biased with an RF bias without DC voltage/current from the pin **225** of the DC input connector leaking into the body **222**. That is, RF and DC are completely isolated from each other through the facility plate **220**.

[0037] In an embodiment, the facility plate **220** may further comprise links **223**. The links **223** may extend up from the body **222**. The links **223** may be a bolt, a screw, a pin, or the like. In some embodiments, the links **223** are electrically conductive. As such, the links **223** may provide electrical coupling between the facility plate **220** and an ESC (not shown in FIG. **2**A). The links **223** may also provide mechanical coupling between the facility plate **220** and the ESC.

[0038] In an embodiment, the facility plate **220** may also comprise a fluidic path **229** between a first surface **227** and a second surface **228**. In the illustrated embodiment, a pair of fluidic paths **229** are shown to enable dual zone gas delivery. Though, a single fluidic path **229** or two or more fluidic paths **229** may be provided in the facility plate **220** in other embodiments. Gas inputs **253** may provide gas to the fluidic paths **229**.

[0039] The fluidic paths **229** in FIG. **2**A have an opening at the first surface **227**, but the opening at the second surface **228** is not visible. This is because the opening at the second surface **228** is out of the plane of FIG. **2**A. As will be described in greater detail below, a horizontal channel may fluidically couple the entrance and exit to the fluid paths **229**. This allows for routing of the fluidic path **229** in order to accommodate existing ESC designs.

[0040] Referring now to FIG. **2**B, a cross-sectional illustration of a cathode assembly **250** is shown, in accordance with an embodiment. In an embodiment, the cathode assembly **250** may comprise a facility plate **220** that is similar to the facility plate **220** in FIG. **2**A and an ESC **200**. In an embodiment, the ESC **200** may be coupled to the facility plate **220** by one or more links **223**. As noted above, the links **223** may provide electrical and mechanical coupling between the ESC **200** and the facility plate **220**. For example, the RF feed line **221** may provide an RF bias to the facility plate **220** that can be transferred to a body **201** of the ESC **200** by the links **223**.

[0041] In an embodiment, the ESC **200** may also comprise a ceramic plate **205**. The ceramic plate **205** may include an embedded electrode **208**. The electrode **208** may be electrically coupled to the DC input connector by a pin **230** that passes through the body **201** of the ESC **200**. In an embodiment, the pin **230** may be surrounded by an electrically insulating collar **231**. This maintains the electrical isolation between the DC bias and the RF bias for the cathode assembly **250**. In an embodiment, the pin **230** may be electrically coupled to the pin **225** in the facility plate **220**. For example, in FIG. **2**B the pin **230** and the pin **225** directly contact each other. In other embodiments, one or more intervening electrically conductive components may be provided between the pin **225** and the pin **230**.

[0042] The ESC **200** may further comprise fluidic channels **215**. The fluidic channels **215** may be provided at a bottom of the ESC **200** and sealed with a lid **216**. The fluidic channels **215** may be suitable for flowing gas and/or liquid within the body **201**. This can be used for cooling or other thermal control of the ESC **200**. In an embodiment, the fluidic channels **215** may be fluidically coupled to the fluidic paths **229** in the facility plate **220** (through a connection out of the plane of FIG. **2**B). In some embodiments, the fluidic channels **215** may comprise a multi-zone configuration. That is, a first fluidic channel **215** may provide an outer zone for backside gas and a

second fluidic channel **215** may provide an inner zone for backside gas.
[0043] Referring now to FIG. **3**A, a zoomed in illustration of a DC input assembly **340** inserted into a hole **333** of an ESC **300** is shown, in accordance with an embodiment. In an embodiment, the DC input assembly **340** may comprise an electrically conductive rod **330** that is electrically coupled to a DC input **332** by an electrically conductive spring **336**. The DC input **332** may be an electrically conductive pad, pin, or the like. The DC input **332** may be electrically coupled to the electrode **308** in the ceramic plate **305** by a via **309** or the like.
[0044] In an embodiment, the rod **330**, the spring **336**, and the DC input **332** may be surrounded by an electrically insulating collar **334**, **335**, and **337** to prevent electrical shorting to the body **301**. The collar **334**, **335**, and **337** is shown as being three distinct parts. In other embodiments, a single electrically insulating component can be used as the collar, or a plurality of components can be coupled together to form the collar. As such, a direct and isolated path from the conductive rod **330** to the electrode **308** can be made in order to apply a DC bias to the electrode **308** for generating a chucking force on a substrate (not shown in FIG. **3**A).
[0045] In an embodiment, the design of the DC input assembly **340** may take into consideration assembly processes. For example, the ESC **300** is often attached to the underlying facility plate (not shown in FIG. **3**A) with a blind install. As such, the ability to provide alignment tolerance is beneficial in order to make assembly easier. In some instances, a portion of the collar **334** may include a recess **331** in order to accommodate a connector for the facility plate. The recess **331** may be a ring shaped recess along the bottom surface of the collar **334** that surrounds the rod **330**.
[0046] The presence of the spring **336** may also be beneficial for the assembly process. Particularly, the spring **336** provides a compressible member that allows for any variation in the placement of the ESC **300** in the Z-dimension to be accommodated. For example, if the ESC **300** is set too “high” the spring **336** expands to provide the proper electrical connection to the electrode **308**. Similarly, if the ESC **300** is set too “low” the spring **336** compresses in order reduce the height of the DC input line.
[0047] Referring now to FIG. **3**B, a cross-sectional illustration of a cathode assembly **350** with an ESC **300** that is coupled to a facility plate **320** is shown, in accordance with an embodiment. In an embodiment, the facility plate **320** is part of the larger pedestal structure on which the ESC **300** is attached. The facility plate **320** may be an interface structure that allows for the DC input assembly **340** and the RF feed **321** to be passed to regions underlying the ESC **300** and out of the chamber (not shown).
[0048] In an embodiment, the facility plate **320** may comprise a body **322** that is a metallic material, such as aluminum or the like. In an embodiment, a hole may pass through the facility plate **320** below the DC input assembly **340**. In an embodiment, a DC input connector may pass through the hole in the facility plate **320**. The DC input connector may comprise an electrically insulating housing or collar **326** that surrounds a pin **325**. The pin **325** may be electrically coupled to the pin **330** of the DC input assembly **340**. In some embodiments, the pin **325** directly contacts the pin **330**. In other embodiments, one or more intervening electrically conductive structures are provided between the pin **325** and the pin **330**.
[0049] In an embodiment, collar **326** may comprise a protrusion **329** at an upper edge of the collar **326**. The protrusion **329** may be a ring-shaped protrusion that surrounds a perimeter of the pin **325**. The protrusion **329** may be sized to insert into the recess **331** of the collar **326** of the DC input assembly **340**. The protrusion **329** and recess **331** interface may be used in order to align the ESC **300** to the facility plate **320** when a blind install or assembly is used.
[0050] In an embodiment, the RF feed **321** passes through the facility plate **320** as well. In other embodiments, the RF feed **321** may be electrically coupled to the body **301** of the ESC **300** through links **323**. Since the pins **325** and **330** are electrically isolated from the conductive bodies **322** and **301**, the DC component and the RF component of the cathode assembly **350** do not interact with each other.

[0051] In an embodiment, the cathode assembly **350** may also be configured to provide a backside gas supply. For example, gas inputs **353** may supply gas to fluidic channels **328** in the facility plate **320**, and the fluidic channels **328** may be fluidically coupled to the fluidic channels **315** that are sealed by a lid **316** in the ESC **300**. The backside gas supply may be separated into two or more zones in some embodiments. For example, an outer zone and an inner zone may be supplied in some embodiments.
[0052] Referring now to FIGS. **4**A-**4**E a series of illustrations depicting a process for forming the fluidic channels within a facility plate **420** is shown, in accordance with an embodiment. In the illustrated embodiment, a single channel is shown for illustrative purposes. However, other embodiments may include two or more channels in order to provide a multi-zone gas distribution system. The channel generated in the facility plate **420** allows for the gas inlet to be routed to a different position in the X-Y plane. This allows for existing gas feed lines and existing ESC designs to be used without the need for redesign. That is, the facility plate **420** reroutes the gas flow path to join a gas feed line to an ESC gas input.
[0053] Referring now to FIG. **4**A, a plan view illustration of a facility plate **420** is shown, in accordance with an embodiment. The facility plate **420** may comprise a body **422**, such as an aluminum body **422**. In FIG. **4**A, only the gas distribution components are shown. Other holes, features, and/or the like are omitted for simplicity. As shown, a first vertical portion **461** and a second vertical portion **462** are formed in the facility plate **420**. The first vertical portion **461** is offset (in the X-Y plane) from the second vertical portion **462**.
[0054] FIG. **4**B is a cross-sectional illustration of a portion of the facility plate **420** along line B-B′ in FIG. **4**A. As shown, the first vertical portion **461** passes into the top surface **428** of the body **422**. However, the first vertical portion **461** does not extend entirely through a thickness of the body **422** in some embodiments. FIG. **4**C is a cross-sectional illustration of a portion of the facility plate **420** along line C-C′ in FIG. **4**A. As shown, the second vertical portion **462** passes into the bottom surface **427** of the body **422**. In some instances, the second vertical portion **462** does not extend entirely through a thickness of the body **422**. In an embodiment, the first vertical portion **461** and the second vertical portion **462** may overlap each other in the Z-dimension.
[0055] Referring now to FIG. **4**D, a plan view illustration of the facility plate **420** after a horizontal channel that fluidically couples the first vertical portion **461** to the second vertical portion **462** is formed is shown, in accordance with an embodiment. In an embodiment, the horizontal channel may comprise a first horizontal portion **463** that extends from an edge of the first vertical portion **461** to an edge of the body **422** and a second horizontal portion **464** that extends from the second vertical portion **462** to the edge of the body **422**. The first horizontal portion **463** and the second horizontal portion **464** may be formed at a similar Z-height so that they intersect at a point **465**. The first horizontal portion **463** and the second horizontal portion **464** may be formed with a drilling operation or the like.
[0056] Referring now to FIG. **4**E, a plan view illustration of the facility plate **420** after plugs **466** and **467** are inserted into the horizontal channel. The plug **466** may be positioned between the intersection point **465** and the edge of the body **422** along the second horizontal portion **464**, and the plug **467** may be positioned between the intersection point **465** and the edge of the body **422** along the first horizontal portion **463**. The plugs **466** and **467** seal the horizontal channel so that the fluidic path proceeds from the first vertical portion **461** to the second vertical portion **462** without leaking out an edge of the body **422**.
[0057] Referring now to FIG. **4**F, a process flow diagram of a process **480** for forming a fluidic path within a facility plate is shown, in accordance with an embodiment. The process **480** may result in a facility plate that is similar in structure to the facility plate **420** shown in FIG. **4**E.
[0058] In an embodiment, the process **480** may begin with operation **481**, which comprises forming a first hole into a first surface of a facility plate. The first hole may be formed with a drilling operation or the like. The first hole may pass partially through a thickness of the facility plate. In an

embodiment, the process **480** may continue with operation **482**, which comprises forming a second hole into a second surface of the facility plate. The second hole may be formed with a drilling operation or the like. The second hole may pass partially through a thickness of the facility plate. In an embodiment, the first hole and the second hole may be offset from each other in the X-Y plane. Additionally, a portion of the first hole may overlap a portion of the second hole in the Z-direction.

[0059] In an embodiment, the process **480** may continue with operation **483**, which comprises forming a first channel from an edge of the facility plate to the first hole. The first channel may be formed with a drilling process or the like. In an embodiment, the process **480** may continue with operation **484**, which comprises forming a second channel from the edge of the facility plate to the second hole. In an embodiment, the second channel intersects the first channel. The second channel may be formed with a drilling process or the like.

[0060] In an embodiment, the process **480** may continue with operation **485**, which comprises plugging the first channel and the second channel. For example, plugs may be provided in the first channel and the second channel between a point of intersection of the channels and the edge of the facility plate. Accordingly, a fluidic path may be provided that starts by going into the first hole, continuing along the first channel, then passing into the second channel, and ultimately out of the second hole.

[0061] Referring now to FIG. **5**, a cross-sectional illustration of a semiconductor processing tool **570** is shown, in accordance with an embodiment. In an embodiment, the semiconductor processing tool **570** may include a plasma processing tool, such as a plasma etching chamber, a plasma dicing chamber, a deposition chamber that uses plasma (e.g., plasma enhanced chemical vapor deposition (PECVD) or plasma enhanced atomic layer deposition (PEALD), etc.), a plasma treatment chamber, or the like.

[0062] In an embodiment, the tool **570** may comprise a chamber **571**. The chamber **571** may be suitable for supporting a vacuum pressure within the chamber **571** in order to support the generation of a plasma **577**. In an embodiment, the chamber **571** may comprise a cathode assembly **550** that is supported over a pedestal **575**. The interior of the pedestal **575** is omitted for simplicity.

[0063] In an embodiment, the cathode assembly **550** may be similar to any of the cathode assemblies described in greater detail herein. For example, the cathode assembly **550** may comprise an ESC **500** that is coupled to a facility plate **520**. In an embodiment, the ESC **500** may comprise a metallic body **501** with a ceramic plate **505** on a top surface of the metallic body **501**. A DC input pin **530** may pass through a hole in the body **501**, and an electrically insulating collar **531** may electrically isolate the DC input pin **530** from the body **501**. The facility plate **520** may also comprise a metallic body **522** with a hole to pass a high voltage DC connector. The high voltage DC connector may comprise a pin **525** surrounded by an electrically insulating collar **526**. The pin **525** may be electrically coupled to the DC input pin **530** of the ESC **500**. The collar **526** electrically isolates the pin **525** from the body **522** of the facility plate **520**.

[0064] In an embodiment, an RF feed **521** may be electrically coupled to the body **522** of the facility plate **520**. The RF bias applied to the body **522** by the RF feed line **521** can be coupled to the body **501** of the ESC **500** through links **523** or the like. Accordingly, the ESC **500** and the facility plate **520** both provide inputs for DC and RF that are electrically isolated from each other. This reduces leakage and can improve tool **570** performance. For example, chucking force applied to a substrate **578** can be reduced compared to existing solutions where the RF and DC inputs are superimposed on each other. In an embodiment, the substrate **578** may be chucked to the ESC **500** through the use of the DC input **530** that is coupled to an electrode (not shown) in the ceramic plate **505**. The substrate **578** may be a wafer (e.g., a silicon wafer) or any other type of substrate used in a semiconductor processing environment.

[0065] In an embodiment, gas feed lines **553** may be provided to the facility plate **520**. The gas feed lines **553** may provide gas to the facility plate **520** and the ESC **500** to enable backside gas delivery for thermal control purposes. In some embodiments, multiple gas feed lines **553** are used to provide multi-zone control. Additionally, the facility plate **520** may include fluidic channels that reroute the gas path to accommodate any input position for gas into the ESC **500**.

[0066] In an embodiment, a showerhead **572** may be provided as a lid to the chamber **571** that is opposite from the ESC **500**. Processing gasses may be flown into the chamber **571** through the showerhead **572**. The showerhead **572** may be biased with RF or microwave frequencies in order to ignite the plasma **577** within the chamber **571**.

[0067] Referring now to FIG. **6**, a process flow diagram of process **690** for processing a substrate with a semiconductor tool that includes a cathode assembly with a DC input that is electrically isolated from an RF input is shown, in accordance with an embodiment. In an embodiment, the process **690** begins with operation **691**, which comprises placing a substrate on a cathode assembly, where the cathode assembly has a DC input that is electrically isolated from an RF input. The cathode assembly used in process **690** may be similar to any of the cathode assemblies described in greater detail herein.

[0068] In an embodiment, the process **690** may continue with operation **692**, which comprises applying a chucking force to the substrate by activating the DC input. Since the DC input is electrically isolated from a remainder of the cathode assembly, the chucking force needed to secure the substrate is reduced compared to existing ESC devices. This also allows for lower current leakage and provides a more efficient tool.

[0069] In an embodiment, the process **690** may continue with operation **693**, which comprises processing the substrate in a plasma environment during an application of an RF bias applied by the RF input. In an embodiment, the processing may include a plasma etching process, a plasma dicing process, or any other treatment process. In an embodiment, the chucking force to the substrate remains substantially constant during the processing of the substrate. Since there is substantially no leakage, the uniform chucking force can be maintained without changing a DC voltage of the DC input.

[0070] In an embodiment, the process **690** may continue with operation **694**, which comprises releasing the chucking force. In an embodiment, the chucking force may be released by reducing a voltage of the DC input. After the chucking force is released, the process **690** may continue with operation **695**. Operation **695** may comprise removing the substrate from the ESC. The substrate may be removed by a wafer handling robot or the like.

[0071] Referring now to FIG. **7**, a block diagram of an exemplary computer system **700** of a processing tool is illustrated in accordance with an embodiment. In an embodiment, computer system **700** is coupled to and controls processing in the processing tool. Computer system **700** may be connected (e.g., networked) to other machines in a Local Area Network (LAN), an intranet, an extranet, or the Internet. Computer system **700** may operate in the capacity of a server or a client machine in a client-server network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. Computer system **700** may be a personal computer (PC), a tablet PC, a set-top box (STB), a Personal Digital Assistant (PDA), a cellular telephone, a web appliance, a server, a network router, switch or bridge, or any machine capable of executing a set of instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while only a single machine is illustrated for computer system **700**, the term “machine” shall also be taken to include any collection of machines (e.g., computers) that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies described herein.

[0072] Computer system **700** may include a computer program product, or software **722**, having a non-transitory machine-readable medium having stored thereon instructions, which may be used to program computer system **700** (or other electronic devices) to perform a process according to embodiments. A machine-readable medium includes any mechanism for storing or transmitting information in a form readable by a machine (e.g., a computer). For example, a machine-readable (e.g., computer-readable) medium includes a machine (e.g., a computer) readable storage medium (e.g., read only memory (“ROM”), random access memory (“RAM”), magnetic disk storage media, optical storage media, flash memory devices, etc.), a machine (e.g., computer) readable transmission medium (electrical, optical, acoustical or other form of propagated signals (e.g., infrared signals, digital signals, etc.)), etc.

[0073] In an embodiment, computer system **700** includes a system processor **702**, a main memory **704** (e.g., read-only memory (ROM), flash memory, dynamic random access memory (DRAM) such as synchronous DRAM (SDRAM) or Rambus DRAM (RDRAM), etc.), a static memory **706** (e.g., flash memory, static random access memory (SRAM), etc.), and a secondary memory **718** (e.g., a data storage device), which communicate with each other via a bus **730**.

[0074] System processor **702** represents one or more general-purpose processing devices such as a microsystem processor, central processing unit, or the like. More particularly, the system processor may be a complex instruction set computing (CISC) microsystem processor, reduced instruction set computing (RISC) microsystem processor, very long instruction word (VLIW) microsystem processor, a system processor implementing other instruction sets, or system processors implementing a combination of instruction sets. System processor **702** may also be one or more special-purpose processing devices such as an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), a digital signal system processor (DSP), network system processor, or the like. System processor **702** is configured to execute the processing logic **726** for performing the operations described herein.

[0075] The computer system **700** may further include a system network interface device **708** for communicating with other devices or machines. The computer system **700** may also include a video display unit **710** (e.g., a liquid crystal display (LCD), a light emitting diode display (LED), or a cathode ray tube (CRT)), an alphanumeric input device **712** (e.g., a keyboard), a cursor control device **714** (e.g., a mouse), and a signal generation device **716** (e.g., a speaker).

[0076] The secondary memory **718** may include a machine-accessible storage medium **731** (or more specifically a computer-readable storage medium) on which is stored one or more sets of instructions (e.g., software **722**) embodying any one or more of the methodologies or functions described herein. The software **722** may also reside, completely or at least partially, within the main memory **704** and/or within the system processor **702** during execution thereof by the computer system **700**, the main memory **704** and the system processor **702** also constituting machine-readable storage media. The software **722** may further be transmitted or received over a network **761** via the system network interface device **708**. In an embodiment, the network interface device **708** may operate using RF coupling, optical coupling, acoustic coupling, or inductive coupling.

[0077] While the machine-accessible storage medium **731** is shown in an exemplary embodiment to be a single medium, the term “machine-readable storage medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) that store the one or more sets of instructions. The term “machine-readable storage medium” shall also be taken to include any medium that is capable of storing or encoding a set of instructions for execution by the machine and that cause the machine to perform any one or more of the methodologies. The term “machine-readable storage medium” shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media.

[0078] In the foregoing specification, specific exemplary embodiments have been described. It will be evident that various modifications may be made thereto without departing from the scope of the following claims. The specification and drawings are, accordingly, to be regarded in an illustrative sense rather than a restrictive sense.

## Claims

**1**. An apparatus, comprising: an electrostatic chuck (ESC) comprising: a first body that is electrically conductive; and a ceramic insert on the first body, wherein an electrode is embedded

within the ceramic insert; and a facility plate coupled to the ESC, wherein the facility plate comprises: a second body that is electrically conductive; a hole through the second body; a DC input connector through the hole; and an RF feed line coupled to the second body, wherein a pin of the DC input connector is electrically isolated from the RF feed line.
**2**. The apparatus of claim 1, wherein the facility plate is electrically coupled to the ESC by one or more links.
**3**. The apparatus of claim 1, wherein the pin of the DC input connector is surrounded by an electrically insulating collar.
**4**. The apparatus of claim 1, wherein the DC input connector is electrically coupled to the electrode by an input assembly that passes at least partially through a thickness of the first body.
**5**. The apparatus of claim 4, wherein the input assembly comprises a spring.
**6**. The apparatus of claim 1, wherein the facility plate comprises a fluidic path from a first surface of the second body to a second surface of the second body, wherein the fluidic path comprises a first vertical portion, a horizontal channel, and a second vertical portion.
**7**. The apparatus of claim 6, wherein the horizontal channel comprises a first branch and a second branch that intersects the first branch.
**8**. The apparatus of claim 1, wherein the ESC comprises a first fluidic channel and a second fluidic channel.
**9**. The apparatus of claim 8, wherein the first fluidic channel is an outer zone backside gas channel, and wherein the second fluidic channel is an inner zone backside gas channel.
**10**. The apparatus of claim 1, wherein a diameter of the facility plate is smaller than a diameter of the ESC.
**11**. An apparatus, comprising: a body with a first surface and a second surface opposite from the first surface, wherein the body is electrically conductive; a hole through the body; an input connector through the hole, wherein the input connector comprises a pin that is electrically conductive and a collar around the pin, wherein the collar is electrically insulating; and an electrically conductive feed line contacting the body.
**12**. The apparatus of claim 11, wherein the feed line is electrically isolated from the pin.
**13**. The apparatus of claim 11, further comprising: a fluidic path from the first surface to the second surface.
**14**. The apparatus of claim 13, wherein the fluidic path comprises a first end on the first surface and a second end on the second surface, and wherein the first end is offset from the second end in an X-Y plane.
**15**. The apparatus of claim 14, wherein the fluidic path comprises a horizontal channel within the body that couples the first end to the second end.
**16**. The apparatus of claim 15, wherein the horizontal channel comprises an intersection point.
**17**. A semiconductor processing tool, comprising: a chamber suitable for maintaining a vacuum environment within the chamber; and a cathode assembly within the chamber, wherein the cathode assembly comprises: an electrostatic chuck (ESC) comprising: a first body that is electrically conductive; and a ceramic insert on the first body, wherein an electrode is embedded within the ceramic insert; and a facility plate coupled to the ESC, wherein the facility plate comprises: a second body that is electrically conductive; a hole through the second body; a DC input connector through the hole; and an RF feed line coupled to the second body, wherein a pin of the DC input connector is electrically isolated from the RF feed line.
**18**. The semiconductor processing tool of claim 17, wherein the semiconductor processing tool is configured for plasma dicing operations.
**19**. The semiconductor processing tool of claim 17, wherein the facility plate comprises a fluidic path with a first vertical portion, a second vertical portion, and a horizontal portion that couples the first vertical portion to the second vertical portion.

**20**. The semiconductor processing tool of claim 17, wherein the DC input connector comprises a pin, and wherein the pin is electrically coupled to the electrode by at least a spring.